Manzil JEE - Practice (2025)

Sequence and Series

Most Important JEE PYQs

Single Correct Type Questions

1. For three positive integers p, q, r, $x^{pq^2} = y^{qr} = z^{p^2r}$ and $r = pq + 1$ such that 3, $3\log_y x$, $3\log_z y$, $7\log_x z$ are in A.P. with common difference $\frac{1}{2}$. Then $r - p - q$ is equal to **[24 Jan, 2023 (Shift-I)]**

(a) 2 (b) 6 (c) 12 (d) –6

2. Let $s_1, s_2, s_3, \ldots, s_{10}$ respectively be the sum to 12 terms of 10 A.P.s whose first terms are 1, 2, 3, ..., 10 and the common differences are 1, 3, 5, ..., 19 respectively. Then $\sum_{i=1}^{10} s_i$ is equal to **[13 April, 2023 (Shift-I)]**

(a) 7380 (b) 7220 (c) 7360 (d) 7260

3. Let S_1 be the sum of first $2n$ terms of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to : **[18 March, 2021 (Shift-II)]**

(a) 5000 (b) 1000 (c) 7000 (d) 3000

4. Let S_n denote the sum of the first n-terms of an arithmetic progression. If $S_{10} = 530, S_5 = 140$, then $S_{20} - S_6$ is equal to : **[22 July 2021 (Shift-II)]**

(a) 1852

(b) 1842

(c) 1872

(d) 1862

5. If $3^{2\sin 2\alpha - 1}$, 14 and $3^{4 - 2\sin 2\alpha}$ are the first three terms of an A.P. for some α, then the sixth term of this A.P. is: **[5 Sep, 2020 (Shift-I)]**

(a) 65 (b) 78 (c) 81 (d) 66

6. The common difference of the A.P. $b_1, b_2, \ldots, b_m$ is 2 more than the common difference of A.P. $a_1, a_2, \ldots, a_n$. If $a_{40} = -159$, $a_{100} = -399$ and $b_{100} = a_{70}$, then b_1 is equal to: **[6 Sep, 2020 (Shift-II)]**

(a) –127 (b) –81 (c) 127 (d) 81

7. The sum of all two digit positive numbers which when divided by 7 yield 2 or 5 as remainder is: **[10 Jan, 2019 (Shift-I)]**

(a) 1256 (b) 1465 (c) 1365 (d) 1356

8. If 19th terms of non-zero A.P. is zero, then its (49th term):(29th term) is: **[11 Jan, 2019 (Shift-II)]**

(a) 4 : 1 (b) 1 : 3 (c) 3 : 1 (d) 2 : 1

9. If sum of the first 21 terms of the series $\log_{9^{\frac{1}{2}}} x + \log_{9^{\frac{1}{3}}} x + \log_{9^{\frac{1}{4}}} x + \ldots$, where $x > 0$ is 504, then x is equal to: **[20 July 2021 (Shift-II)]**

(a) 7 (b) 9 (c) 243 (d) 81

10. Let $a_1, a_2, \ldots a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{2i-1}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to **[9 Jan, 2019 (Shift-I)]**

(a) 52

(b) 57

(c) 47

(d) 42

11. $\frac{1}{3^2 - 1} + \frac{1}{5^2 - 1} + \frac{1}{7^2 - 1} + \ldots + \frac{1}{(201)^2 - 1}$ is equal to **(2021)**

(a) $\frac{101}{404}$ (b) $\frac{25}{101}$

(c) $\frac{101}{408}$ (d) $\frac{99}{400}$

12. Let α and β be the roots of $x^2 - 3x + p = 0$ and γ and δ be the roots of $x^2 - 6x + q = 0$. If α, β, γ, δ form a geometric progression. Then ratio $(2q + p) : (2q - p)$ is: **[4 Sep, 2020 (Shift-I)]**

(*a*) 3 : 1 (*b*) 5 : 3 (*c*) 9 : 7 (*d*) 33 : 31

13. Let *a*, *b*, *c*, *d* and *p* be any non zero distinct real numbers such that $(a^2 + b^2 + c^2) p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) = 0$. Then: **[6 Sep, 2020 (Shift-I)]**

(*a*) *a*, *c*, *p* are in G. P. (*b*) *a*, *b*, *c*, *d* are in A. P.

(*c*) *a*, *c*, *p* are in A.P. (*d*) *a*, *b*, *c*, *d* are in G.P.

14. If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to: **[12 April, 2019 (Shift-II)]**

(*a*) βγ (*b*) 0 (*c*) αγ (*d*) αβ

15. In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4th, 6th and 8th terms is equal to: **[26 Feb, 2021 (Shift-I)]**

(*a*) 30 (*b*) 32 (*c*) 26 (*d*) 35

16. The sum to 10 terms of the series

$$\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \ldots \text{ is}$$

[1 Feb, 2023 (Shift-I)]

(*a*) $\frac{59}{111}$ (*b*) $\frac{55}{111}$ (*c*) $\frac{56}{111}$ (*d*) $\frac{58}{111}$

17. Let A_1 and A_2 be two arithmetic means and G_1, G_2, G_3 be three geometric means of two distinct positive numbers. The $G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$ is equal to **[15 April, 2023 (Shift-I)]**

(*a*) $2(A_1 + A_2) G_1 G_2$ (*b*) $(A_1 + A_2)^2 G_1 G_3$

(*c*) $(A_1 + A_2) G_1^2 G_3^2$ (*d*) $2(A_1 + A_2) G_1^2 G_3^2$

18. If $S_n = 4 + 11 + 21 + 34 + 50 + \ldots$ to *n* terms, then $\frac{1}{60}(S_{29} - S_9)$ is equal to **[10 April, 2023 (Shift-II)]**

(*a*) 226 (*b*) 220 (*c*) 223 (*d*) 227

19. If gcd $(m,n) = 1$ and $1^2 - 2^2 + 3^2 - 4^2 + \ldots + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$, then $m^2 - n^2$ is equal to **[6 April, 2023 (Shift-II)]**

(*a*) 200

(*b*) 240

(*c*) 220

(*d*) 180

20. Let $x,y > 0$. If $x^3y^2 = 2^{15}$, then the least value of $3x + 2y$ is **[24 June, 2022 (Shift-II)]**

(*a*) 30 (*b*) 32 (*c*) 36 (*d*) 40

21. If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}, x > 0$ is 14, then the value of α is equal to **[28 July, 2022 (Shift-I)]**

(*a*) 32 (*b*) 64 (*c*) 128 (*d*) 256

22. If the sum of the first 20 terms of the series $\log_{(7^{1/2})} x + \log_{(7^{1/3})} x + \log_{(7^{1/4})} x + \ldots$ is 460, then *x* is equal to. **[5 Sep, 2020 (Shift-II)]**

(*a*) 7^2

(*b*) e^2

(*c*) $7^{1/2}$

(*d*) $7^{46/21}$

Integer Type Questions

23. The 8th common term of the series

$S_1 = 3 + 7 + 11 + 15 + 19 + \ldots$

$S_2 = 1 + 6 + 11 + 16 + 21 + \ldots$ is

[30 Jan, 2023 (Shift-II)]

24. The sum of the common terms of the following three arithmetic progressions. **[1 Feb, 2023 (Shift-II)]**

$3, 7, 11, 15, \ldots\ldots\ldots\ldots, 399$

$2, 5, 8, 11, \ldots\ldots\ldots\ldots 359$ and

$2, 7, 12, 17, \ldots\ldots, 197$, is equal to ________.

25. Let 3, 6, 9, 12, ... upto 78 terms and 5, 9, 13, 17, ... upto 59 terms be two series. Then, the sum of the terms common to both the series is equal to. **[29 June, 2022 (Shift-II)]**

26. Let $a_1, a_2, \ldots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then

$$12\left(\frac{1}{\sqrt{a_{10}} + \sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}} + \sqrt{a_{12}}} + \ldots \frac{1}{\sqrt{a_{17}} + \sqrt{a_{18}}}\right) \text{ is equal to}$$

[31 Jan, 2023 (Shift-I)]

27. Let $a_1, a_2, a_3, \ldots$ be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ is equal to **[29 Jan, 2023 (Shift-I)]**

28. Let $0 < z < y < x$ be three real numbers such that $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in an arithmetic progression and $x, \sqrt{2}y, z$ are in a geometric progression. If $xy + yz + zx = \frac{3}{\sqrt{2}}xyz$, then $3(x + y + z)^2$ is equal to____ **[8 April, 2023 (Shift-II)]**

29. The remainder on dividing $1 + 3 + 3^2 + 3^3 + ... + 3^{2021}$ by 50 is……….. **[24 June, 2022 (Shift-II)]**

30. If the sum of the series $\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2.3}+\frac{1}{3^2}\right)+.$

$$\left(\frac{1}{2^3}-\frac{1}{2^2\cdot 3}+\frac{1}{2\cdot 3^2}-\frac{1}{3^3}\right)+$$

$$\left(\frac{1}{2^4}-\frac{1}{2^3\cdot 3}+\frac{1}{2^2\cdot 3^2}-\frac{1}{2.3^3}+\frac{1}{3^4}\right)+....\text{is}\frac{\alpha}{\beta},\text{where } \alpha \text{ and}$$

β are co-prime, then $\alpha + 3\beta$ is equal to....

[15 April, 2023 (Shift-I)]